

Daily Algebra Drill #7

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Weighted AM–GM · Vieta jumping · Power mean inequality · Infinite products via telescoping.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

No expansion. Recognise the pattern, write the answer.

- Factorise: $x^5 - x$.
- Given $ab + bc + ca = 4$ and $a + b + c = 3$, find $(a + b + c)^2 - (a^2 + b^2 + c^2)$.
- Without a calculator: $7^2 - 5^2 + 3^2 - 1^2$.
- The roots of $x^2 - 8x + 3 = 0$ are p and q . Find $\left(p - \frac{1}{p}\right)\left(q - \frac{1}{q}\right)$.
- Simplify: $\frac{(2n)!}{(2n-1)!} - \frac{n!}{(n-1)!}$.
- Factorise: $27x^3 - 8$.
- Given $t = x - \frac{1}{x}$, write $t^2 + 4$ in terms of x .
- Evaluate: $\log_2 8 + \log_2 4 - \log_2 32$. (*No log rules needed: convert to powers of 2.*)
- Given $f(x) = 2x + 1$ and $g(x) = x^2 - 1$, find $f(g(x)) - g(f(x))$.
- Simplify: $\frac{a^2 - b^2}{a - b} - \frac{a^2 - c^2}{a - c}$.

Section B Manipulation Drills

(≤50 s each)

Fractions, structure, identities. Minimum working.

- Evaluate the telescoping product: $\prod_{k=2}^n \left(1 - \frac{1}{k}\right)$ and find its value when $n = 100$.
- Simplify: $\frac{x^2 + 3x - 10}{x^2 - x - 2} \cdot \frac{x^2 + x - 2}{x^2 + 6x + 5}$.

3. Find the value of $a^3 + b^3 + c^3$ given $a + b + c = 2$, $a^2 + b^2 + c^2 = 6$, $abc = -2$.
4. The expression $p(x) = x^4 + 4$ can be written as a product of two quadratic factors with integer coefficients. Find them. (*This is Sophie Germain: add and subtract $4x^2$.*)
5. Given $f(x) = \frac{2x+1}{x-1}$, find $f^{-1}(x)$ and verify $f(f^{-1}(x)) = x$.
6. Prove: the product of four consecutive integers is always one less than a perfect square. (*Let the integers be $n-1, n, n+1, n+2$ and pair them.*)
7. Simplify: $\frac{1}{1+x} + \frac{1}{1-x} + \frac{2}{1+x^2} + \frac{4}{1+x^4}$.
8. The equation $x^3 + px + q = 0$ has roots α, β, γ . Express $\alpha^3 + \beta^3 + \gamma^3$ in terms of p and q without using Newton's identity. (*Each root satisfies $r^3 = -pr - q$. Add.*)
9. Prove that if $a + b + c = 0$ then $(a^2 + b^2 + c^2)^2 = 2(a^4 + b^4 + c^4)$.
10. Solve: $\frac{x+1}{x-2} + \frac{x-2}{x+1} = \frac{10}{3}$.

Section C Substitution & Structure

(≤90 s each)

One clean substitution is worth ten lines of algebra.

1. Using weighted AM–GM ($\lambda a + (1-\lambda)b \geq a^\lambda b^{1-\lambda}$ for $\lambda \in (0, 1)$), prove that $\frac{x}{3} + \frac{2y}{3} \geq x^{1/3}y^{2/3}$ for $x, y > 0$.
2. Find all real pairs (x, y) satisfying: $x + y = 1$ and $x^4 + y^4 = \frac{7}{8}$.
3. Prove: for $a, b > 0$,

$$\sqrt{\frac{a^2 + b^2}{2}} \geq \frac{a+b}{2} \geq \sqrt{ab} \geq \frac{2ab}{a+b}.$$
 (*Prove each $\geq$ from right to left using a single algebraic step each.*)
4. Solve: $27^x + 27^{1-x} = 28$.
5. For real a, b, c with $a + b + c = 1$ and $abc > 0$, prove $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$.
6. The sum $S = 1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$. Verify for $n = 3$, then use it to evaluate $3^3 + 4^3 + \cdots + 10^3$ without computing each cube.
7. Find all real solutions to $|x^2 - 4| = |x - 2| \cdot |x + 2|$.
8. Given $x, y > 0$ with $x + y = 4$, find the maximum of x^2y .

Section D Challenge

(SMC / BMO1 difficulty)

No brute force. Each question rewards one key observation.

1. Vieta Jumping. Suppose a and b are positive integers with $a^2 + b^2 = k(ab + 1)$ for some positive integer k .

(a) Show $k = 4$ is achieved by $(a, b) = (8, 2)$.

(b) Assume a solution (a, b) with $a \geq b$ exists. Consider a as a root of $x^2 - kbx + (b^2 - k) = 0$ and let a' be the other root. Show $a' = kb - a$ and $a'b = b^2 - k$. (*This is the essence of Vieta jumping: the other root is smaller.*)

2. Find all polynomials $p(x)$ with real coefficients such that $(x - 1)p(x + 1) = (x + 2)p(x)$ for all real x .

3. For positive reals a, b, c with $a + b + c = 1$, find the minimum of

$$\frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c}.$$

4. Evaluate:

$$\prod_{n=2}^{\infty} \frac{n^3 - 1}{n^3 + 1}.$$

(Factor each term as a ratio of products using sum and difference of cubes. The result telescopes.)

5. Find all integer solutions to $x^3 + y^3 = (x + y)^2$.

“Do not worry about your difficulties in mathematics. I assure you mine are still greater.” — Albert Einstein

other prep to look at today:
Daily Integral for Calculus and Logic